

QUICK START GUIDE

Forward Discrete Fourier Transform HP15c CE software pac

(rev2)

Pepin Torres, P.E.
pepin(dot)torres{at}gmail(dot)com

Introduction:

This software pac implements an N-point forward Discrete Fourier Transform for any real-valued matrix in row-vector form when stored in `Matrix A` (e.g., $\mathbb{R}^{1 \times 15}$).

See https://en.wikipedia.org/wiki/Discrete_Fourier_transform for technical details of math involved.

Who is this for?

This is for anyone who is ever in dire need of calculating the Forward DFT of a *real-valued* sequence with no computer or advanced calculator in sight. This software pac can run with the calculator in default mode or 15.2 mode (see *Size and Timing Considerations* section for details).

Excel file includes listing, comments, stack states, sanity check example, and timing table.

Note: Program is short enough that inputting by hand is not objectionable plus the added benefit of not having to blow away existing programs. The .mem file is provided for those who would rather not type it in.

To calculate N-point Forward DFT:

Prerequisites:

Stack: N/A

Registers: None

Matrices: `Matrix A` of size $[1 \times N]$ containing a real-valued sequence

where $N = \{1 \dots 29\}$ for default mode (max N is assuming no other data in memory)
 $N = \{1 \dots 77\}$ for 15.2 mode (max N is assuming no other data in memory)

Usage:

Press **GSB B** (state of stack does not matter)

Output:

`Matrix B` of size $[(N/2+1) \times 2]$

where column 1 contains the real part of the result
column 2 contains the imaginary part of the result

Program calculates the positive half of the spectrum with the negative half implied thanks to the [conjugate symmetry property](#).

Registers used: R0, R1, R.0, R.1, R.2, R.4, R.5

BONUS: To only calculate the k^{th} -point of the N-point Forward DFT:

Prerequisites:

Stack: k in the x-register

where $k = \{0 \dots N-1\}$

Registers: None

Matrices: Matrix A of size $[1 \ N]$ containing a real-valued sequence

where $N = \{1 \dots 99\}$ for default mode (max N is assuming no other data in memory)

$N = \{1 \dots 99\}$ for 15.2 mode (max N is assuming no other data in memory)

Usage:

Press **GSB A** (with k in x-register)

Output:

Value of Forward DFT at bin k is stored in $R.4$ (real part) and $R.5$ (imaginary part)

Registers used: $R0, R1, R.0, R.1, R.2, R.4, R.5$

Size and Timing Considerations:

Program size: 80 instructions, 83 bytes

Max DFT size: 29 points in default mode, 77 points in 15.2 mode
(assuming no other data or programs in memory.)

Execution Time vs Number of Points

